

Hyodae Seo

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BIO

Dr. Hyodae Seo is an Associate Professor in the Department of Oceanography and an Associate Director of the Uehiro Center for the Advancement of Oceanography. He is a physical oceanographer and climate scientist with a broad range of research and teaching interests in ocean and atmospheric processes and their coupled interactions, with a focus on weather and climate. He combines high-resolution coupled modeling with theories of geophysical fluid dynamics and analyses of in situ and satellite observations to better understand ocean-atmosphere boundary layer processes, improve their representation in numerical models, and evaluate their influence on the ocean, weather, and climate. Dr. Seo actively contributes to developing sustainable ocean-observing strategies and strives to engage the public on issues related to the ocean, extreme weather, climate, and ocean-based renewable energy research.

EDUCATION

2007: Ph.D. in Oceanography, Scripps Institution of Oceanography, UC San Diego

Thesis title: [Mesoscale coupled ocean-atmosphere interaction](#)

Thesis advisors: [Dr. Arthur J. Miller](#) and Dr. John Roads ([deceased](#))

2002: B.S. in Meteorology, Yonsei University, Seoul, South Korea

POSITIONS HELD

2024 – : Associate Professor, Department of Oceanography, University of Hawai'i at Mānoa

2023 – 2024: Senior Scientist, Woods Hole Oceanographic Institution (WHOI)

2018 – 2023: Associate Scientist with Tenure, WHOI

2014 – 2018: Associate Scientist, WHOI

2010 – 2014: Assistant Scientist, WHOI

2009 – 2010: Visiting Assistant Researcher, IPRC, University of Hawai'i at Mānoa

2007 – 2008: NOAA Climate and Global Change Postdoctoral Fellow, UCLA & UH Mānoa

2007: Visiting Scientist, International Pacific Research Center, Univ. Hawai'i at Mānoa

2002 – 2007: Graduate Research Assistant, Scripps Institution of Oceanography, UCSD

AWARDS AND HONORS

- WHOI Francis E. Fowler IV Ocean and Climate Fellow (2022, [link](#))
- Office of Naval Research (ONR) Young Investigator Award (2015, [WHOI press release](#))
- NOAA Climate and Global Change Postdoctoral Fellowship (2007-2009, [class 17](#))
- Physical Oceanography Dissertation Symposium V, Honolulu (2008)
- NCAR Advanced Study Program (ASP) Graduate Student Visiting Fellowship (2006)
- Scripps Institution of Oceanography Frieman Director's Prize for Excellence in Graduate Student Research (2006, [UCSD press release](#))

- Outstanding Student Paper Award, Ocean Sciences Meeting, Honolulu (2006)

PUBLICATIONS (*led by student and postdoc under my direct supervision)

Under Review, Revision, or In Press:

- Song, H., H. Seunu, **H. Seo**, D. J. McGillicuddy, Jr., K. Kwak, J. Yun, H.-J. Kim, J. Lee, K. Park, and J. Park, Physical and Biogeochemical Characteristics of Four Types of Eddies in the Southern Ocean, *Geophys. Res. Lett.*, *Submitted*
- Kirinchch, A., and coauthors, including **H. Seo**, and C. Sauvage, Improving the Understanding and Forecasting of Winds over the Northeast U.S. Shelf: The Third Wind Forecast Improvement Project (WFIP3). *Bull. Amer. Meteor. Soc.*, *Submitted*
- *Renkl, C., **H. Seo**, and A. J. Miller, Marine heatwave impacts on landfalling atmospheric river on the U.S. West Coast. *Submitted*
- *Kim, Y.-J., K. Tanaka, **H. Seo**, K. Komatsum, Y. Matsumura, C. Sauvage, and C. Renkl, A Study on the Role of Time- and Spatially Varying Wind in Coastal Circulation Using High-resolution Oceanic and Atmospheric Numerical Simulations. *J. Geophys. Res. Oceans*, *Submitted*
- Renault, L., J. Wenegrat, **H. Seo**, P. Marchesiello, The Energetics of Fine-Scale Air-Sea Interactions. *Annual Review of Marine Science*, *In Press*

Published:

2026

69. Cho, A., H. Song, I.-J. Moon, **H. Seo**, R. Sun, M. R. Mazloff, A. C. Subramanian, A. J. Miller, 2026: Modulation of Tropical Cyclone Intensity by Current-Wind Interaction, *npj Clim. Atmos. Sci.*, 9, 44, <https://doi.org/10.1038/s41612-025-01316-1>

2025

68. *Sauvage, C., **H. Seo**, S. Zippel, C.-A. Clayson, and J. B. Edson, 2025: Fetch-dependent Surface Wave Responses To Offshore Wind Farms in the Northeast U.S. Coast, *J. Geophys. Res. Oceans*, 130, e2025JC023156, <https://doi.org/10.1029/2025JC023156>
67. **Seo, H.**, C. Sauvage, C. Renkl, J. K. Lundquist, and A. Kirincich, 2025: Sea Surface Warming and Ocean-to-Atmosphere Feedback Driven by Large-Scale Offshore Wind Farms Under Seasonally Stratified Conditions. *Sci. Adv.*, 11, eadw7603, <https://www.science.org/doi/10.1126/sciadv.adw7603>
66. Raj, A., B. P. Kumar, V. Jampana, P. G. Remya, **H. Seo**, N. Sureshkumar, and P. R. Rao, 2025: Wind-Stress Misalignment in the Presence of Swells in the Bay of Bengal. *Frontiers in Marine Science*, 12. <https://doi.org/10.3389/fmars.2025.1647849>
65. Yang, S., H. Bae, M.-S. Park, M. Bourassa, C. C. Nam, S. Cocke, D. W. Shin, B. W. Barr, **H. Seo**, D.-H. Cha, M.-H. Kwon, D. Kim, K.-Y. Jung, and B.-M. Kim, 2025: Sea spray effects on typhoon prediction in the Yellow and East China Seas: Case studies using a coupled atmosphere-ocean-wave model for Lingling (2019) and Maysak (2020). *Env. Res. Lett.*, 20, 054028. <https://doi.org/10.1088/1748-9326/adc616>
64. Cho, A., H. Song, **H. Seo**, R. Sun, M. R. Mazloff, A. C. Subramanian, B. D. Cornuelle, and A. J. Miller, 2025: Dynamic and Thermodynamic coupling between the Atmosphere and Ocean near the Kuroshio Current and Extension System. *Ocean Modell.*, 194, 102496. <https://doi.org/10.1016/j.ocemod.2024.102496>

2024

63. *Sauvage, C., **H. Seo**, B. W. Barr, J. B. Edson, and C. A. Clayson, 2024: Misaligned Wind-Waves Behind Atmospheric Cold Fronts. *J. Geophys. Res. Oceans*, 129, e2024JC021162. <https://doi.org/10.1029/2024JC021162>

62. Shinoda, T., T. G. Jensen, Z. Lachkar, Y. Masumoto, and **H. Seo**, 2024: Modeling the Indian Ocean, Intraseasonal variability in the Indian Ocean region. The Indian Ocean and its role in the global climate system. Elsevier Publishing, Editors: C. C. Ummenhofer and R. R. Hood. <https://doi.org/10.1016/B978-0-12-822698-8.00013>
- 2023
61. *Steffen, J. D., **H. Seo**, C. A. Clayson, S. Pei, and T. Shionda, 2023: Impacts of Tidal Mixing on Diurnal to Intraseasonal Air-Sea Interactions in the Maritime Continent. *Deep-Sea Res.-II*, 212, 105343. <https://doi.org/10.1016/j.dsr2.2023.105343>
60. *Castillo-Trujillo A. C., Y.-O. Kwon, P. Fratantoni, K. Chen, **H. Seo**, M. A. Alexander, and V. S. Saba, 2023: An evaluation of eight global ocean reanalyses for the Northeast U.S. continental shelf. *Prog. Oceanogr.*, <https://doi.org/10.1016/j.pocean.2023.103126>
59. Clayson, C. A., C. A. DeMott, S. P. de Szoeke, P. Chang, G. R. Foltz, R. Krishnamurthy, T. Lee, A. Molod, D. G. Ortiz-Suslow, J. Pullen, D. H. Richter, **H. Seo**, P. C. Taylor, E. Thompson, B. V. Bôas, C. J. Zappa, and P. Zuidema, 2023: A New Paradigm for Observing and Modeling of Air-Sea Interactions to Advance Earth System Prediction. A US CLIVAR Report, US CLIVAR Project Office, 86 pp. <https://doi.org/10.5065/24j7-w583>.
58. Beaudin, E., E. Di Lorenzo, A. J. Miller, **H. Seo**, and Y. Joh, 2023: Impact of Extratropical Northeast Pacific SST on U.S. West Coast Precipitation, *Geophys. Res. Lett.*, 50, e2022GL102354. <https://doi.org/10.1029/2022GL102354>
57. *Sauvage, C., **H. Seo**, C. A. Clayson, and J. B. Edson, 2023: Improving wave-based air-sea momentum flux parameterization in mixed seas. *J. Geophys. Res. Oceans*, 128, e2022JC019277. <https://doi.org/10.1029/2022JC019277>
56. **Seo, H.**, L. W. O'Neill, M. A. Bourassa, A. Czaja, K. Drushka, B. Fox-Kemper, I. Frenger, S. T. Gille, B. P. Kirtman, S. Minobe, A. G. Pendergrass, L. Renault, M. J. Roberts, N. Schneider, R. J. Small, A. Stoffelen, and Q. Wang, 2023: Ocean Mesoscale and Frontal-scale Ocean-Atmosphere Interactions and Influence on Large-scale Climate: A Review. *J. Climate*, 36, 1981-2013. <https://doi.org/10.1175/JCLI-D-21-0982.1>
- 2021
55. Kwak, K. H. Song, J. Marshall, **H. Seo**, and D. McGillicuddy, Jr., 2021: Suppressed pCO₂ in the Southern Ocean due to the interaction between current and wind. *J. Geophys. Res. Oceans*, 126, e2021JC017884. <https://doi.org/10.1029/2021JC017884>
54. Shinoda, T., S. Pei, W. Wang, J. X. Fu, R.-C. Lien, **H. Seo**, and A. Soloviev, 2021: Climate Process Team: improvement of ocean component of NOAA Climate Forecast System relevant to Madden-Julian Oscillation simulations. *J. Adv. Model. Earth Syst.*, 13, e2021MS002658. <https://doi.org/10.1029/2021MS002658>
53. **Seo, H.**, H. Song, L. W. O'Neill, M. R. Mazloff, and B. D. Cornuelle, 2021: Impacts of ocean currents on the South Indian Ocean extratropical storm track through the relative wind effect. *J. Climate*, 34, 9093-9113. <https://doi.org/10.1175/JCLI-D-21-0142>
52. Pei, S., T. Shinoda, J. Steffen, and **H. Seo**, 2021: Substantial Sea Surface Temperature cooling in the Banda Sea associated with the Madden-Julian Oscillation in the boreal winter of 2015. *J. Geophys. Res. Oceans.*, 126, e2021JC017226. <https://doi.org/10.1029/2021JC017226>
51. Shroyer, E., and Co-authors, including **H. Seo**, 2021: Bay of Bengal Intraseasonal Oscillation and the 2018 Monsoon Onset. *Bull. Amer. Meteor. Soc.*, 102(10), E1936-E1951. <https://doi.org/10.1175/BAMS-D-20-0113>

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 49. Sun, R., A. C. Subramanian, B. D. Cornuelle, M. R. Mazloff, A. J. Miller, F. M. Ralph, **H. Seo**, and I. Hoteit, 2021: The role of air-sea interactions in atmospheric river events: Case studies using the SKRIPS regional coupled model. *J. Geophys. Res. Atmosphere*, 126, e2020JD032885. <https://doi.org/10.1029/2020JD032885>
- 2020
48. Liang, Y.-C., M.-H. Lo, C.-W. Lan, **H. Seo**, C. C. Ummenhofer, S. Yeager, R.-J. Wu, and J. D. Steffen, 2020: Amplified Seasonal Cycle in Hydroclimate over the Amazon River Basin and its Plume Region in the Atlantic. *Nat. Commun.*, 11, 4390. <https://doi.org/10.1038/s41467-020-18187-0>
 47. *Yuan, X., C. C. Ummenhofer, **H. Seo**, and Z. Su, 2020: Relative contributions of heat flux and wind stress on the spatiotemporal upper-ocean variability in the tropical Indian Ocean. *Env. Res. Lett.*, 15, 084047. <https://doi.org/10.1088/1748-9326/ab9f7f>
 46. Gentemann, C., C. A. Clayson, S. Brown, T. Lee, R. Parfitt, J. T. Farrar, M. Bourassa, P. J. Minnett, **H. Seo**, S. T. Gille, and V. Zlotnicki, 2020: FluxSat: Measuring the ocean-atmosphere turbulent exchange of heat and moisture from space. *Remote Sensing*, 12, 1796. <https://doi.org/10.3390/rs12111796>
 45. Sprintall, J., K. A. Reed, A. H. Butler, G. R. Foltz, S. G. Penny, and **H. Seo**, 2020: Best Practice Strategies for Process Studies designed to Improve Climate Modeling, *Bull. Amer. Met. Soc.*, 101, E1842-E1850. <https://doi.org/10.1175/BAMS-D-19-0263>
 44. Song, H., J. Marshall, D. J. McGillicuddy Jr., and **H. Seo**, 2020: The impact of the current-wind interaction on the vertical processes in the Southern Ocean. *J. Geophys. Res.-Oceans*, 125, e2020JC016046. <https://doi.org/10.1029/2020JC016046>
 43. Gopalakrishnan, G., A. C. Subramanian, A. J. Miller, **H. Seo**, and D. Sengupta, 2020: Estimation and prediction of the upper ocean circulation in the Bay of Bengal, *Deep-Sea Res. II*, 172, 104721. <https://doi.org/10.1016/j.dsr2.2019.104721>
 42. Hagos, S., G. R. Foltz, C. Zhang, E. Thompson, **H. Seo**, S. Chen, A. Capotondi, K. A. Reed, C. DeMott, and A. Protat, 2020: Atmospheric Convection and Air-Sea Interactions over the Tropical Oceans: Scientific Progress, Challenges and Opportunities. *Bull. Amer. Met. Soc.*, 101, E253–E258. <https://doi.org/10.1175/BAMS-D-19-026>
 41. Kwon, Y.-O., **H. Seo**, C. C. Ummenhofer, and T. M. Joyce, 2020: Impact of Multidecadal Variability in Atlantic SST on Winter Atmospheric Blocking. *J. Climate*, 33, 867–892. <https://doi.org/10.1175/JCLI-D-19-0324>
- 2019
40. **Seo, H.**, A. C. Subramanian, H. Song, and J. S. Chowdary, 2019: Coupled effects of ocean current on wind stress in the Bay of Bengal: Eddy energetics and upper ocean stratification. *Deep-Sea Res. II*, 168, 104617. <https://doi.org/10.1016/j.dsr2.2018.09.007>
 39. *Prend, C. J., **H. Seo**, R. A. Weller, and J. T. Farrar, 2019: Impact of freshwater plumes on intraseasonal upper ocean variability in the Bay of Bengal. *Deep-Sea Res.-II*, 161, 63–71. <https://doi.org/10.1029/2018GL0810>
 38. Joyce, T. M., Y.-O. Kwon, **H. Seo**, and C. C. Ummenhofer, 2019: Meridional Gulf Stream shifts can influence wintertime variability in the North Atlantic Storm Track and

- Greenland Blocking. *Geophys. Res. Lett.*, 46, 1702-1708. <https://doi.org/10.1175/JCLI-D-18-0413.1>
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- 2018**
36. *Parfitt, R., and **H. Seo**, 2018: A New Framework for Near-Surface Wind Convergence over the Kuroshio Extension and Gulf Stream in Wintertime: The Role of Atmospheric Fronts. *Geophys. Res. Lett.*, 45, 9909–9918. <https://doi.org/10.1175/JCLI-D-18-0413.1>
35. Pullen, J., R. Allard, **H. Seo**, A. J. Miller, S. Chen, L. P. Pezzi, T. Smith, P. Chu, J. Alves, and R. Caldeira, 2018: Coupled ocean-atmosphere forecasting at short and medium time scales. *J. Mar. Res.*, 75, 877-921. https://elischolar.library.yale.edu/journal_of_marine_research/4
34. *Jin, X, Y.-O. Kwon, C. C. Ummenhofer, **H. Seo**, Y. Kosaka, and J. S. Wright, 2018: Distinct mechanisms of decadal subsurface heat content variations in the eastern and western Indian Ocean modulated by tropical Pacific SST, *J. Climate*, 31, 7751-7769. <https://doi.org/10.1175/JCLI-D-18-0184.1>
33. *Jin, X., Y.-O. Kwon, C. C. Ummenhofer, **H. Seo**, F. U. Schwarzkopf, A. Biastoch, C. W. Böning, and J. S. Wright, 2018: Influences of Pacific Climate Variability on Decadal Subsurface Ocean Heat Content Variations in the Indian Ocean. *J. Climate*, 31, 4157-4174. <https://doi.org/10.1175/JCLI-D-17-0654.1>
32. Kwon, Y.-O., A. Camacho, C. Martinez-Zayas, and **H. Seo**, 2018: North Atlantic Eddy-driven Jet and Blocking Variability in the Community Earth System Model Version 1 Large Ensemble Simulations. *Climate Dyn.*, 51, 3275-3289. <https://doi.org/10.1007/s00382-018-4078-6>
- 2017**
31. **Seo, H.**, Y.-O. Kwon, T. M. Joyce, and C. C. Ummenhofer, 2017: On the predominant nonlinear response of the extratropical atmosphere to meridional shift of the Gulf Stream. *J. Climate*, 30, 9679-9702. <https://doi.org/10.1175/JCLI-D-16-0707.1>
30. Morrow, R., L.-L. Fu, T. Farrar, **H. Seo**, P.-Y. Le Traon, 2017: Ocean eddies and mesoscale variability. *Satellite Altimetry Over Oceans and Land Surfaces*, In D. Stammer and A. Cazenave, editors, *Satellite Altimetry Over Ocean and Land Surfaces*. CRC Press, Taylor and Francis Group, [Satellite Altimetry Over Oceans and Land Surfaces](#).
29. Centurioni, L. R., and Co-authors including **H. Seo**, 2017: Northern Arabian Sea Circulation-Autonomous Research (NASCar): A Research Initiative Based on Autonomous Sensors. *Oceanography*, 30, 74-87. <https://doi.org/10.5670/oceanog.2017.224>
28. Ummenhofer, C. C., **H. Seo**, Y.-O. Kwon, R. Parfitt, S. Brands, and T. M. Joyce, 2017: Emerging European winter precipitation pattern linked to atmospheric circulation changes over the North Atlantic region in recent decades. *Geophys. Res. Lett.*, 44, 8557–8566. <https://doi.org/10.1002/2017GL074188>
27. **Seo, H.**, 2017: Distinct influence of air-sea interactions mediated by mesoscale sea surface temperature and surface current in the Arabian Sea. *J. Climate*, 30, 8061-8079. <https://doi.org/10.1175/JCLI-D-16-0834.1>
26. Miller, A. J., M. Collins, S. Gualdi, T. G. Jensen, V. Misra, L. P. Pezzi, D. W. Pierce, D. Putrasahan, **H. Seo**, and Y.-H. Tseng, 2017: Coupled Ocean-Atmosphere-Hydrology

- Modeling and Predictions, *J. Mar. Res.*, 75, 361-402. https://elischolar.library.yale.edu/journal_of_marine_research/437
25. *Parfitt, R., A. Czaja, and **H. Seo**, 2017: A simple diagnostic for the detection of atmospheric fronts, *Geophys. Res. Lett.*, 44, 4351–4358. <https://doi.org/10.1002/2017GL073662>
 24. Jongaramrungruang, S., **H. Seo**, and C. C. Ummenhofer, 2017: Intraseasonal rainfall variability in the Bay of Bengal during the Summer Monsoon: Coupling with the ocean and modulation by the Indian Ocean Dipole. *Atmos. Sci. Lett.*, 18, 88–95. <https://doi.org/10.1002/asl.729>
- 2016**
23. Chowdary, J. S., G. Srinivas, T. S. Fousiya, A. Parekh, C. Gnanaseelan, **H. Seo**, and J. A. MacKinnon, 2016: Representation of Bay of Bengal Upper-Ocean Salinity in General Circulation Models. *Oceanography*, 29, 38–49. <https://doi.org/10.5670/oceanog.2016.37>
 22. **Seo, H.**, A. J. Miller, and J. R. Norris, 2016: Eddy-wind interaction in the California Current System: dynamics and impacts, *J. Phys. Oceanogr.*, 46, 439-459. <https://doi.org/10.1175/JPO-D-15-0086.1>
 21. Brink, K. H. and **H. Seo**, 2016: Continental Shelf Baroclinic Instability 2: Oscillating wind forcing, *J. Phys. Oceanogr.*, 46, 569-582. <https://doi.org/10.1175/JPO-D-15-0048.1>
- 2015**
20. *Oltmanns, M., F. Straneo, **H. Seo**, and G. W. K. Moore, 2015: The role of wave dynamics and small-scale topography for downslope wind events in southeast Greenland. *J. Atmos. Sci.*, 72, 2786-2805. <https://doi.org/10.1175/JAS-D-14-0257.1>
 19. Cavanaugh, N., T. Allen, A. Subramanian, B. Mapes, **H. Seo**, A. J. Miller, 2015: The Skill of Tropical Linear Inverse Models in Hindcasting the Madden-Julian Oscillation. *Climate Dyn.*, 44, 897-906. <https://doi.org/10.1007/s00382-014-2181-x>
- 2014**
18. **Seo, H.**, A. C. Subramanian, A. J. Miller, and N. R. Cavanaugh, 2014: Coupled impacts of the diurnal cycle of sea surface temperature on the Madden-Julian Oscillation. *J. Climate*, 27, 8422-8443. <https://doi.org/10.1175/JCLI-D-14-00141.1>
 17. Park, J.-Y., J.-S. Kug, **H. Seo**, and J. Bader, 2014: Impact of bio-physical feedbacks on the tropical climate in coupled and uncoupled GCMs. *Climate Dyn.*, 43, 1811-1827. <https://doi.org/10.1007/s00382-013-2009-0>
 16. **Seo, H.**, Y.-O. Kwon, and J.-J. Park, 2014: On the effect of the East/Japan Sea SST variability on the North Pacific atmospheric circulation in a regional climate model. *J. Geophys. Res.-Atmospheres*, 119, 418-444. <https://doi.org/10.1002/2013JD020523>
 15. Subramanian, A., M. Jochum, A. J. Miller, R. Neale, **H. Seo**, D. Waliser, and R. Murtugudde, 2014: The MJO and Global warming: A study in CCSM4. *Climate Dyn.*, 42, 2019-2031. <https://doi.org/10.1007/s00382-013-1846-1>
- 2013**
14. **Seo, H.**, and J. Yang, 2013: Dynamical response of the Arctic atmospheric boundary layer process to uncertainties in sea ice concentration. *J. Geophys. Res-Atmosphere*, 118, 12383-12402. <https://doi.org/10.1002/2013JD020312>
 13. Putrasahan, D. A. J. Miller, and **H. Seo**, 2013: Regional coupled ocean-atmosphere downscaling in the Southeast Pacific: Impacts on upwelling, mesoscale air-sea fluxes, and ocean eddies. *Ocean Dynm.*, 63, 463-488. <https://doi.org/10.1007/s10236-013-0608-2>

12. **Seo, H.** and S.-P. Xie, 2013: Impact of ocean warm layer thickness on the intensity of Hurricane Katrina in a regional coupled model. *Meteor. Atmos. Phys.*, 122, 19-32. <https://doi.org/10.1007/s00703-013-0275-3>
11. Putrasahan, D. A. J. Miller, and **H. Seo**, 2013: Isolating Mesoscale Coupled Ocean-Atmosphere Interactions in the Kuroshio Extension Region. *Dyn. Atmos. Oceans*, 63, 60-78. <https://doi.org/10.1016/j.dynatmoce.2013.04.001>
- 2012
10. **Seo, H.**, K. H. Brink, C. E. Dorman, D. Koracin, and C. A. Edwards, 2012: What determines the spatial pattern in summer upwelling trends on the U.S. West Coast? *J. Geophys. Res.-Oceans*, 117, C08012. <https://doi.org/10.1029/2012JC008016>
9. Alexander, M.A., **H. Seo**, S.-P. Xie, and J.D. Scott, 2012: ENSO's impact on the gap wind regions of the eastern tropical Pacific Ocean. *J. Climate*, 25, 3549-3565. <https://doi.org/10.1175/JCLI-D-11-00320.1>
- 2011
8. **Seo, H.**, and S.-P. Xie, 2011: Response and Impact of Equatorial Ocean Dynamics and Tropical Instability Waves in the Tropical Atlantic under Global Warming: A regional coupled downscaling study. *J. Geophys. Res.-Oceans*, 116, C03026. <https://doi.org/10.1029/2010JC006670>
- 2009
7. **Seo, H.**, S.-P. Xie, R. Murtugudde, M. Jochum, and A. J. Miller, 2009: Seasonal effects of Indian Ocean freshwater forcing in a regional coupled model. *J. Climate*, 22, 6577-6596. <https://doi.org/10.1175/2009JCLI2990>
- 2008
6. **Seo, H.**, R. Murtugudde, M. Jochum, and A. J. Miller, 2008: Modeling of Mesoscale Coupled Ocean-Atmosphere Interaction and its Feedback to Ocean in the Western Arabian Sea. *Ocean Modell.*, 25, 120-131. <https://doi.org/10.1016/j.ocemod.2008.07.003>
5. **Seo, H.**, M. Jochum, R. Murtugudde, A. J. Miller, and J. O. Roads, 2008: Precipitation from African Easterly Waves in a Coupled Model of the Tropical Atlantic. *J. Climate*, 21, 1417-1431. <https://doi.org/10.1175/2007JCLI1906.1>
4. Small, R. J., S. de Szoeke, S. P. Xie, L. O'Neill, **H. Seo**, Q. Song, P. Cornillon, M. Spall, and S. Minobe, 2008: Air-Sea Interaction over Ocean Fronts and Eddies. *Dyn. Atmos. Oceans*, 45, 274-319. <https://doi.org/10.1016/j.dynatmoce.2008.01.001>
- 2007
3. **Seo, H.**, M. Jochum, R. Murtugudde, A. J. Miller, and J. O. Roads, 2007: Feedback of Tropical Instability Wave - induced Atmospheric Variability onto the Ocean. *J. Climate*, 20, 5842-5855. <https://doi.org/10.1175/JCLI4330.1>
2. **Seo, H.**, A. J. Miller and J. O. Roads, 2007: The Scripps Coupled Ocean-Atmosphere Regional (SCOAR) model, with applications in the eastern Pacific sector. *J. Climate*, 20, 381-402. <https://doi.org/10.1175/JCLI4016.1>
- 2006
1. **Seo, H.**, M. Jochum, R. Murtugudde and A. J. Miller, 2006: Effect of Ocean Mesoscale Variability on the Mean State of Tropical Atlantic Climate. *Geophys. Res. Lett.*, 33, L09606. <https://doi.org/10.1029/2005GL025651>

TEACHING

@ UHM (since August 2024)

- Spring 2026: OCN/ATMO 665: Small-Scale Air-Sea Interactions
- Spring 2025, OCN 760: Topics in Physical Oceanography (Air-Sea Interaction and

Offshore Wind)

- Spring 2025, OCN780: Ocean Seminar
- Fall 2024, GES 311: The Changing Earth and Climate System. Guest lecture, “Why offshore wind energy: Necessity, Challenges and Oceanographers’ Role”, Nov 21
- @ MIT/WHOI Joint Program (until July 2024)
- Spring Semester, 2023: Air-Sea Interaction and Boundary Layers, MIT/WHOI Joint Program Course 12.870. Co-teaching with Tom Farrar
- Fall Semester, 2017, 2019, 2021: Climate Variability and Diagnostics, MIT/WHOI Joint Program Course 12.860. Co-teaching with Caroline Ummenhofer
- Fall Semester, 2016, 2017, 2018, 2019: Introduction to (Observational) Physical Oceanography, MIT/WHOI Joint Program Course 12.808. Co-taught with John Toole

ADVISING AND MENTORING

Thesis Committee

@ UHM (since August 2024)

- M.S. Yuta Norden (University of Hawai'i, Oceanography), 2026- Present
- Ph.D. Meghan Bradley (University of Hawai'i, Oceanography), 2025- Present
- Ph.D. Ryo Dobashi (University of Hawai'i, Oceanography), 2025 — 2026
- Ph.D. Ajin Cho (Yonsei University), 2024— 2026
- Comprehensive Exam Committee, Carla Balzeau, Oceanography, UHM (2025)
- Comprehensive Exam Committee, Vera Stockmayer, Oceanography, UHM (2025)

@ MIT/WHOI Joint Program (until July 2024)

- Ph.D. General Exam Advisor for Anthony Meza (MIT-WHOI Joint Program), 2022-2023
- Ph.D. Thesis Committee for Hanyuan Liu (MIT-WHOI Joint Program), 2021-2023
- Ph.D. External Examiner for Christoph Renkl (Dalhousie University, Oceanography), 2020
- Ph.D. Thesis committee for Jared Buckley (Univ. Massachusetts, Dartmouth), 2015-2019
- M.S. Thesis committee for Xin Zhou (Florida Institute of Technology), 2015-2016
- M.S. Thesis committee for Sara Bosshart (MIT-WHOI Joint Program), 2013

Graduate Advisor

@ UHM (since August 2024)

- Rachel Wyn Pauly: Ph.D. student, Oceanography Department, UHM (2025–present)
- Clint Reyes: Summer graduate student, ORE, UHM (2025)

Research Supervisor

@ UHM (since August 2024)

- Dr. César Sauvage: Oceanographic Researcher (2024/11–present)
- Dr. Sid Kerhalkhar: Postdoc (2025/09–present)
- Allison Ho (2026–)

@ WHOI (until July 2024)

- Shikhar Rai (2023–)
- Benjamin Barr (2023–)
- Christoph Renkl (2023-2025) at WHOI. Now an Assistant Professor, University of Bonn
- Alma Carolina Castillo-Trujillo (2021-2024).
- César Sauvage (2020-2023), now RA3 at WHOI and UH
- John Steffen (2019-2021, now at NOAA)
- Rhys Parfitt (2016-2018, now Associate Professor, FSU)

Visiting Students and Scholars

@ UHM (since August 2024)

- Prof. Changhyun Yoo: Professor, Ewha Womans University, South Korea (2025–2026)

@ WHOI (until July 2024)

- Yoo-Jun Kim (University of Tokyo), Nov. 2023–Mar. 2024: High-resolution wind modeling.
- Peisen Tan (University of Miami), Summer of 2023: Laboratory and numerical modeling of short wind waves and swell interactions.
- Sam Bartusek (Princeton University, SSF 2019, now at Columbia University): Coastal Ocean impacts on landfalling ARs ([Bartusek et al. 2021](#)).
- Yuqing Liu (Bremen Univ., Guest Student, 2018, now at AWI, Germany): Submesoscale air-sea coupling
- Manthan Shah (Northeastern Univ., Guest Student, 2018): Scale-dependence of SST-wind coupling
- Xu Yuan (Univ. Twente, Guest Student, 2018): Upper-ocean variability in the tropical Indian Ocean ([Yuan et al. 2020](#))
- Xiaolin Jin (Tsinghua University, Visiting Student, 2017), Decadal Subsurface Ocean Heat Content Variations in the Indian Ocean. ([Jin et al. 2018a](#), [Jin et al. 2018b](#))
- Samuel Coakley (Rutgers University, SSF 2017): Southeast Asian monsoon variability in the CESM-Last Millennium Ensemble
- David Smith (Northeastern University, Guest Student, 2016): Decadal predictability of the North Atlantic Blocking
- Channing Prend (Columbia Univ., SSF 2016, Scripps Ph.D.): Impact of freshwater plumes on intraseasonal SST variability in the Bay of Bengal ([Prend et al. 2018](#), [Weller et al. 2019](#))
- Siraput Jongaramrungruang (Univ. Cambridge, SSF 2015, now at Caltech): Interannual modulation of intraseasonal convection in the Bay of Bengal [Jongaramrungruang et al. 2017](#))
- Alicia Camacho (Valparaiso University, UCAR SOARS student 2015, Ph. D. from Stony Brook University): North Atlantic Oscillation, Jet and Blocking in CESM1 Large Ensemble Simulations. ([Kwon et al. 2018](#))
- Carlos Martinez (Texas A&M University, UCAR SOARS student 2014, Ph.D. from Columbia University): North Atlantic Atmospheric Blocking and Atlantic Multi-decadal Oscillation in CESM1 Large Ensemble Simulations. ([Kwon et al. 2018](#))

PROFESSIONAL ACTIVITIES**Community Activities****@ UHM (since August 2024)**

- UCAR President's Advisory Committee on University Relations (PACUR): 2023–Present ([link](#))
- Associate Editor, Journal of Climate (2019– Present)

@ WHOI (until July 2024)

- WHOI Representative for UCAR (2018–2023)
- Co-Chair, US CLIVAR Working Group on Mesoscale and Frontal-scale Ocean-Atmosphere Interactions and Influence on Large-scale Climate (2019–Present, [link](#))
- Member, US CLIVAR Air-Sea Transition Zone Study Group (2022–2023, [link](#))
- (Past) Member, AMS STAC Committee on Air-Sea Interaction (2019– 2022)
- (Past) Member, US CLIVAR Process Study and Model Improvement Panel (2016–2020)
- (Past) Associate Editor: Atmospheric Science Letters (2015–2017)
- (Past) Member, AMS STAC Committee on Coastal Environment (2011–2017)

Committee and Service**@ UHM (since August 2024)**

- Department of Oceanography Awards Committee (2025- Present)

@ WHOI (until July 2024)

- WHOI Physical Oceanography Recruitment Committee for Scientific Staff (2020-2021)
- WHOI High-Performance Computing Strategy Committee (2022-2023)
- WHOI Summer Student Fellowship Coordinator (2017-2019)
- WHOI HPC User Advisory Committee (2010-2017)
- WHOI Physical Oceanography Department seminar series organizer (2010-2011)

Agency Proposal Review Panels

- NASA SWOT (2012); DOE ESMD (2020)

Society Memberships

- American Geophysical Union (2002-Present)
- American Meteorological Society (2002-Present)
- The Oceanography Society (2012-Present)

FUNDED RESEARCH (+ denotes Seo is the Lead or Sole PI)**@ UHM (since August 2024): Raised >\$1.69M as lead PI**

- **+ONR (current, \$524,481):** Improving Air-Sea Flux Parameterizations to Better Understand the Diabatic Effects of Ocean and Surface Waves in Atmospheric Rivers.
- **+ONR (current, \$632,333):** Improving the model simulation of surface wave impacts on air-sea fluxes, turbulent boundary layers, and their impacts on Indian monsoons in the Arabian Sea.
- **+WHOI (NASA) Subcontract (current, \$314,183):** Improving coupled atmosphere-ocean processes in NU-WRF for the simulation of coast-threatening extratropical cyclones in the northeastern US.
- **+WHOI (DOE) Subcontract (current, \$87,582):** Improving High Resolution Offshore Wind Resource Assessments and Forecasts using Observations in the MA/RI Lease Areas.
- **+WHOI (NSF) Subcontract (current, \$139,693):** Improving understanding of coupled impacts of oceans and waves on air-sea fluxes in the US Northeast Coast.

@ WHOI (until July 2024) External Grants: Raised ~\$6.36M as lead PI, contributed to \$11.49M as co-PI

- **#ONR (current, \$50,000):** Improving the model simulation of surface wave impacts on air-sea fluxes, turbulent boundary layers, and their impacts on Indian monsoons in the Arabian Sea.
- **#NOAA (current, \$773,753):** Exploiting coupled ocean-atmosphere-wave model simulations to identify observational requirements for air-sea interaction studies across the tropical Pacific. Co-PI: Wijffels
- **#NSF (current, \$831,828):** Improving understanding of coupled impacts of oceans and waves on air-sea fluxes in the US Northeast Coast. Co-PI: Sauvage, Clayson, and Edson. (\$139,693 subcontracted to UHM).
- **#NASA (current, \$1,166,106):** Improving coupled atmosphere-ocean processes in NU-WRF for the simulation of coast-threatening extratropical cyclones in the northeastern US. (\$314,183 subcontract to UHM).
- **DOE (current, \$10M):** Improving High Resolution Offshore Wind Resource Assessments and Forecasts using Observations in the MA/RI Lease Areas. co-PI, PI: Kirincich (WHOI). (\$87,582 subcontracted to UHM)

- **#NSF (current, \$497,273):** Collaborative Research: Coupled Ocean-Atmosphere Feedbacks Affecting California Coastal Climate: Current Conditions and Future Projections.
 - **NOAA (current, \$658,442):** Regional multi-year prediction for the Northeast U.S. Continental Shelf. Co-PI. PI: Kwon (WHOI).
 - **#NOAA (2019-2024, \$767,746):** Coupled Ocean-atmosphere interaction mediated by ocean mesoscale eddies in the Northwestern Tropical Atlantic Ocean. Co-PI: Clayson (WHOI)
 - **#ONR (2017-2023, \$505,185):** Coupled Air-sea Interactions in the Bay of Bengal: Dynamics and Predictability of Monsoon Intraseasonal Oscillation.
 - **NOAA (2017-2022, \$267,797):** Upper ocean processes in the Maritime Continent and their impact on the air-sea interaction and MJO predictability, co-PI. Lead PI: Shinoda (TAMUCC)
 - **NSF (2014-2019, \$670,121):** Quantifying Predictability Limits, Uncertainties, Mechanisms, and Regional Impacts of Pacific Decadal Climate Variability, Institutional PI. Lead PI: Miller (SIO)
 - **NOAA (2016-2019, \$180,000):** Improvement of MJO simulation in NCEP Coupled Forecast System: Upper ocean and air-sea coupled processes, Institutional PI. Lead PI: Shionda (TAMUCC)
 - **NSF (2014-2019, \$500,000):** Decadal Variability in the North Atlantic Extra-Tropics: The Role of Coupling Between Atmospheric Blocking and the Atlantic Multidecadal Oscillation, co-PI. Lead PI: Ummenhofer (WHOI)
 - **#ONR Young Investigator Award (2015-2019, \$490,482):** Dynamics, Impacts, and Predictability of Coupled Ocean-Atmosphere Interactions in the Northern Indian Ocean.
 - **NASA (2015-2018, \$494,210):** Coupling Between the Atmospheric Intra-Seasonal Variability and Ocean Circulation in the Northern Hemisphere, co-I. PI: Kwon (WHOI)
 - **#NSF (2014-2015, \$72,000):** Regional Coupled Ocean-Atmosphere Feedback Processes Affecting Climate Along the California Coast.
 - **#ONR (2011-2014, \$91,000):** Coupled Ocean-Atmosphere Dynamics and Predictability of MJO's.
- © **WHOI (until July 2024) Internal (competitive) Grants, Secured ~\$495K**
- **#Fowler Ocean and Climate Fellow (2021-2023, \$140,213):** Western boundary currents and air-sea interactions: toward skillful predictive understanding and modeling of extreme storm and short-term climate events
 - **#Independent Study Award (2019-2021, \$64,000):** Southern Ocean Eddies and Atmospheric Storm Tracks
 - **#Ocean and Climate Change Institute (2015-2017, \$55,000):** A regional coupled model with explicit convection: diurnal rainfall and air-sea-land interaction in the Maritime Continent.
 - **#Independent Study Award (2013-2015, \$49,000):** Submesoscale air-sea Interactions.
 - **#Ocean and Climate Change Institute (2011-2013, \$140,000):** Dynamical analysis of surface wind responses to sea ice and surface temperature variations in the Arctic Ocean: Synthesis of data and model simulation. Co-PI: Yang (WHOI)
 - **#Independent Study Award (2012-2013, \$47,000):** Trends in coastal upwelling and sea breeze in the California-Oregon Coast: Land-ocean-atmosphere interactions in data and model.